Unit 4

Campbell Biology, AP Edition, 12e

Unit 4

Mechanisms of Evolution

Cassandra Extavour

Cassandra Extavour is Professor in the Department of Organismic and Evolutionary Biology at Harvard University, where she and her students are studying genes that control early embryonic development, as well as the evolutionary origins of these genes and how their functions have changed over time. Born in Toronto, Canada, Dr. Extavour received a B.Sc. (honors) in Molecular Biology and Genetics from the University of Toronto, followed by a Ph.D. in the same subject area from Universidad Autónoma de Madrid, Spain. After completing two postdoctoral fellowships (one in Greece, the other in the United Kingdom), she worked for

several years as a Research Associate at Cambridge University, UK, before joining the faculty at Harvard. Dr. Extavour is a dedicated advocate for women and underrepresented groups in science, and she also has performed worldwide as a professional soprano.

An Interview with: Cassandra Extavour

How did you first become interested in science?

Although my first love was music, and music remains an essential part of my life, toward the end of high school I also became quite interested in biology. My conversations with a friend who was fascinated by human behavior led me to wonder how the brain causes people to exhibit different behaviors. But how could I learn about that? My parents did not go to college. I didn't even know that there were careers in science that were not in medicine. I thought the only way to learn about the brain was to be a doctor. So I entered university as a pre-med. But by the time I was finishing my undergraduate studies, I had learned about research labs. I got a summer internship in a developmental genetics lab, and I loved it—it was like entering a new world where I could figure out the answers to questions that interested me.

What are the main questions you are asking in your research?

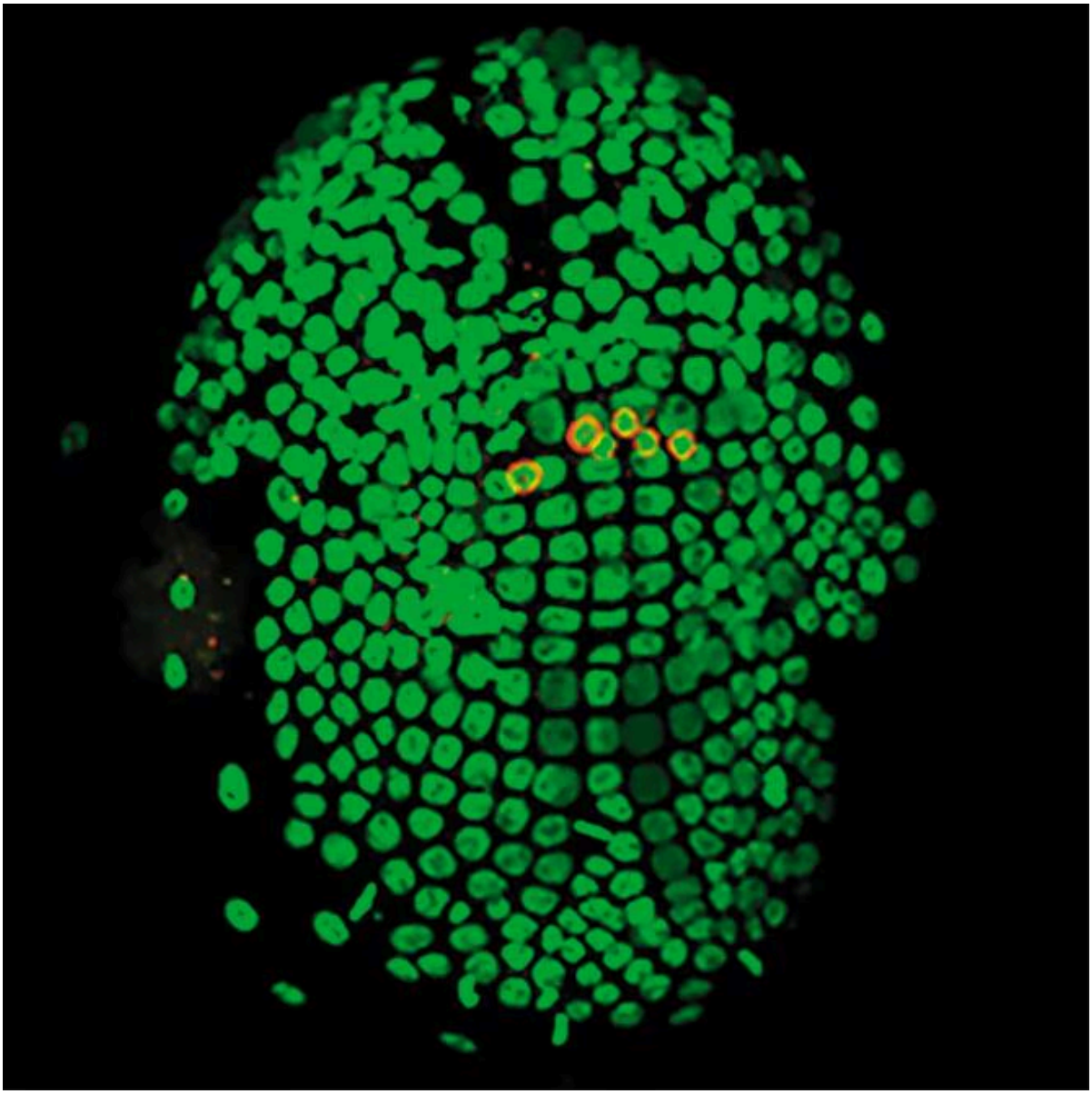
I'm interested in understanding the genes that control embryonic development and how the function of those genes has changed in different organisms due to evolution. Take a human, for example. Our bodies have many different types of cells. The instructions for everything that each of our cells needs to do are contained in the cell's DNA. All of our cells have exactly same DNA. What's mysterious is, since every cell in the body has the same set of instructions, why aren't they all doing the same thing? I work mostly with insects, but the same question applies to them (and to all multicellular organisms). We're trying to figure out how the same set of genes is used in different ways by different cells. We're also interested in where these genes came from in the first place and how they have evolved over time.

> ***"What's mysterious is, since every cell in the body has the same set of instructions, why aren't they all doing the same thing?"***

Much of your research has been on "evo-devo." Say more about what that means.

Evo-devo stands for evolutionary developmental biology. Scientists in this field try to understand the difference in development between, let's say, a coconut and a rose. They're both plants, they both make seeds, they both make pollen, but they don't look the same at all. Why is that? Or the fact that, when you compare the development of a monkey and a human, you can't detect many differences in their embryos or their genes. How can they be similar in so many ways and yet end up as such different organisms? In my lab, we are particularly interested in egg and sperm cells—known as germ cells. We're trying to figure out how an

embryo determines which cells become germ cells and how it tells those cells, “Okay, listen: When the animal you are part of becomes an adult, you will need to make eggs and sperm.” We’ve found that in some cases, the same genes are used by a variety of different species to make germ cells. But in other cases, we’ve been surprised to find that a gene used to produce germ cells in one organism is used to make a completely different type of cell in another closely related organism.



Embryo of the crustacean Parhyale hawaiensis showing cells (in red) that will develop into germ cells

What do you find most rewarding about your job?

When I was in high school, I never would have thought that I could get paid to think of interesting questions and then try to answer them. I love doing that, and I love the amazing freedom that comes with my job—I get to decide what I’m going to do each day. But the best part is working with students as they gain the

independence, knowledge, and confidence to listen to my suggestions and then say, “I’m not sure that’s the best experiment. Here’s an experiment that I think would be better, and here’s why.” That is a great moment. What they are really saying is, “I’m not learning to be a scientist anymore—I am a scientist.” They’ve come to understand what scientists do: We make the best proposal we can, we test it as best we can, and we say, “This is what I think the results from my experiments mean.”